SeoYeon Kim

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Profile Summary

- MSc Data Science graduate with experience as a Data Scientist in a UK insurance company and 2+ years at a digital therapeutics startup, passionate about building impactful solutions addressing real-world challenges.
- **Right to work in the UK: NO VISA SPONSORSHIP REQUIRED.**

Education

University of Bristol — Bristol, United Kingdom

Master of Science, Major in Data Science — *Sep 2025 – Oct 2026*

- *Key Modules*: Statistical Computing and Empirical Methods, Large-Scale Data Engineering, Introduction to AI and Text Analytics, Visual Analytics, Data Science Methods and Practice (average mark: 76%, Distinction)

Yonsei University — Seoul, Korea

Bachelor of Science in Engineering — *Mar 2018 – Aug 2024*

- *Double Majors*: Bio-Convergence and Engineering & Electrical and Electronic Engineering
- *Dissertation*: Literature review on natural compounds targeting estrogen receptor-positive breast cancer using 3D pharmacophore/Quantitative Structure–Activity Relationship (QSAR) models (Jun 2022).

Professional Experience

Allianz UK — Bristol, United Kingdom

Data Scientist, Claims Operations Department, Full-time, Internship — *June – Oct 2026*

- *Data Science Dissertation Project | Feedback-Loop Detection and Mitigation Application*
  - Led an individual project on the Claims Operations vehicle total loss prediction service, where the Machine Learning (ML) loop itself forced total-loss labels with no human agent involved.
  - Defined the stakeholders and business cost of the production ML system, and collaborated with the Claims team through daily stand-ups to translate and link research methods to its data, decisions, and constraints, setting objectives and the success metric around the business's precision requirement.
  - Implemented complex academic concepts as a four-layer Python framework, built on the Strategy design pattern with a SHAP-based loop-severity estimator, that can be applied to the team's production ML system.
  - Quantified a self-fulfilling prophecy loop across model versions with a regression discontinuity-style comparison near the decision cutoff (causal inference), reducing the loop signature by 52% while precision increased from 98.52% to 98.89%.
  - Integrated model repositories with incompatible XGBoost versions into one `uv`-managed project through a single configuration layer.
  - Documented each experiment run's settings as JSON metadata beside pipeline outputs, making results reproducible and traceable.
  - Presented the application to the Claims team, where its applicability to other ML systems was recognised.

180 Degrees Consulting — Bristol, United Kingdom

Data Scientist, Grapple Law Project, Contract — *Feb – Apr 2026*

- World's largest university-based consultancy, connecting student volunteers with social enterprises.
- *n8n-Based Automated Pipeline | Automated Legal Platform User Retention Pipeline*
  - Built an n8n nudge pipeline that segments users by an 11-level case progression hierarchy and triggers LLM-personalised retention emails through SendGrid.
  - Tested hypotheses on Grapple data with statistical tests, identifying three factors linked to settlement offers: employer acknowledgement, documents sent, and email volume.
  - Surfaced 31 potential nudges with the team, then implemented 3 n8n workflows with LLM prompt engineering to drive conversion to Grapple's subscription: failed email resending, an example legal letter for returning users, and a personalised email by legal progression.
  - Pulled user data from Supabase with SQL inside the n8n workflows and processed it with TypeScript in the following nodes.
  - Built a user engagement dashboard with a Sankey diagram showing progression across the 11-level legal hierarchy toward settlement offers.

WELT Corporation — Seoul, Korea

Software Engineer, Development Department, Full Time — *Apr 2024 – Jul 2025*

- Series C-funded digital therapeutics (DTx) company with CE-marked insomnia DTx, expanding to European markets.
- *TypeScript Backend Development | Digital Therapeutic (DTx) for Insomnia (Germany, Medical Device Class IIa)*
  - Architected and deployed backend infrastructure for a German clinical trial DTx application in 3 months, launching in August 2025 with zero errors in user testing.
  - Implemented Command Query Responsibility Segregation and event-driven architecture in a monorepo, ensuring modularity and no circular dependencies.
  - Automated cloud provisioning and deployment with Terraform, Terragrunt, Docker, and GitHub CI/CD on Google Cloud Platform (GCP).
  - Built a type-safe Prisma data layer for PostgreSQL and integrated Redis for caching and sessions.
  - Designed API contracts with frontend developers through real-time discussions.
- *TypeScript Backend Development | AI-Powered Healthcare Chatbot*
  - Prototyped a LangGraph-based Slack AI chatbot with role-based access control to automate prescription code generation.
- *Python Backend Development & MLOps | Automated Sleep Pattern Prediction Service*
  - Owned the development of an automated ML pipeline using Random Forest Regression (pandas, scikit-learn) to predict sleep patterns from iOS HealthKit data, demonstrating the feasibility of personalised sleep insights.
  - Automated model retraining with Google Cloud Run Service and Job.
  - Implemented a custom sleep duration algorithm, enabling feature engineering without wearable device data.
  - Managed an 8-week project lifecycle and presented findings to 18 stakeholders across Development, Clinical, and Product teams with an interactive Q&A.
- *iOS Development | On-Device AI Sleep Prediction Application*
  - Analysed underutilised iOS HealthKit data and proposed AI-powered features to improve DTx patients' treatment outcomes.
  - Developed an on-device AI sleep prediction application using Random Forest Regression in CoreML.
  - Presented ML concepts to 6 clinical staff, including the Chief Medical Officer (CMO), enabling collaborative development of AI-driven feature requirements.

Software Engineer Intern, Development Department, Full Time *Sep 2023 – Mar 2024*

- *Python Backend Development | Digital Therapeutic (DTx) for Insomnia*
  - Redesigned backend logic for the DTx application with a context-flow-based AI agentic system, designing and developing 34 features to improve therapeutic effectiveness and user engagement.
  - Facilitated Series C fundraising, securing £7.5M from Korean Development Bank, IMM Investment, Shinhan Venture Investment, and Handok (Jun 2024).
  - Presented AI agent architecture to 30+ company-wide stakeholders, including C-suite executives, facilitating knowledge sharing on AI agentic system adoption.
- *Quality Assurance | Digital Therapeutic (DTx) for Eating Disorders*
  - Led Quality Assurance testing and patient data monitoring for a DTx, supporting regulatory approval.

Regulatory Affairs Intern, Clinical Department, Full Time *Jun – Aug 2023*

- Drafted Software Architecture and Detail Design documents for Class IIa Software as Medical Device.

Relevant Projects

- "Serverless AI Image-Processing Architecture Design and Auto-Scaling Optimisation of a Distributed Text-Processing Service on Amazon Web Services (AWS)"
  Relevant skills: serverless architecture design on AWS (Lambda, SageMaker, API Gateway), distributed data processing and auto-scaling (SQS, EC2 Auto Scaling, DynamoDB), load testing (processing time for 120 files cut by 76%)
- "Image-to-Sentence Retrieval via CNN Embedding Regression: A Multi-Dataset Comparative Study"
  Relevant skills: computer vision (CNN image embeddings), image-to-sentence retrieval, evaluation metric design, team leadership (Jira, GitHub code review)
- "A Decision-Support Dashboard for Local Authority Integration Officers: Housing Insecurity and Economic Integration of Migrants, UK 2021"
  Relevant skills: Bayesian forecasting, PCA, K-Means, UMAP, Tableau dashboard design, user testing
- "How Far Can We Trust a Deep Probabilistic Time Series Forecast? — Investigating Calibration Degradation in DeepAR and Temporal Fusion Transformer (TFT), and its Post-hoc Conformal Correction via EnbPI"
  Relevant skills: time series forecasting, prediction uncertainty and conformal correction

- Allianz UK × Barnardo's Hackathon 2026: Selected as 1 of 2 winning teams out of 6 by clustering donors with K-Means on engineered activity features, recommending retention actions, and presenting the expected donation increase as the monetary impact.
- London AI×Bio Hackathon 2026: Placed 2nd of 15 teams (WilbeLABS, NVIDIA, OpenAI) with an agentic workflow using GPT-Rosalind, NVIDIA BioNeMo, DiffDock, and OpenFold to identify minimal point mutations conferring herbicide resistance while preserving protein structure.
- Lloyds × UoB Datathon 2026: Benchmarked 11 ML models for fraud detection and designed a performance-vs-carbon leaderboard.
- Wise × HuddleHive Women in Tech Hackathon 2026: Built and deployed AdLens, an undisclosed-advertisement detector for Instagram posts that increases transparency for customers, using fine-tuned DistilBERT (1.6M posts, disclosure hashtags masked) and Gemini-generated explanations in a FastAPI/Next.js app.
- Member of Bristol Data Science Society (BDSS) (Sep 2025 – Sep 2026), attending weekly workshops.
- Headstart Global Launchpad EMEA 2026: Led the Lumenia team in the Healthcare Tech case competition, delivering a fully evidenced B2B mental health platform pitch within six days.
- PLUS Award: Awarded for 70+ hours of extra work experience, employability workshops, and mock interviews, with a reflective report.
- International PGT Mentoring: Selected as 1 of 55 mentees from 226 applicants for a 6-month programme on UK work culture and careers.
- Climate Fresk Workshop: Completed a 3-hour collaborative climate workshop, mapping interconnected climate drivers and effects.

Laboratory Experience & Academic Activities

Yonsei University IBS CNM & NanoBME — Sep 2022 – Jun 2023

Team Leader of Precision Medicine | NANOSEA (NanoScience & Engineering Academia) — *Seoul, Korea*

Intern of the Gene Editing Team (Mini Science Factory), May 2023 — *Seoul, Korea*

Yonsei University Bioelectronics and Electrical Engineering Laboratory BLISS — Jan – May 2023

Undergraduate Thesis Research Intern | Conductive Hydrogel Development for Soft Bioelectronics — *Seoul, Korea*

Yonsei University Medicinal Food Engineering Laboratory — Mar – Jun 2022

Undergraduate Thesis Research Intern | Literature Review on Computational Approaches for Breast Cancer Drug Discovery — *Seoul, Korea*

Skills

Programming & Query Languages: Python, SQL, TypeScript
Machine Learning & Analytics: scikit-learn, pandas, XGBoost, SHAP, CoreML, LangGraph, DistilBERT, Random Forest Regression, K-Means, PCA, UMAP, Bayesian Ridge, hypothesis testing (Mann-Whitney U, chi-squared, logistic regression)
Cloud & MLOps: Amazon Web Services (AWS: Lambda, SageMaker, S3, API Gateway, SQS, EC2 Auto Scaling, DynamoDB), Google Cloud Platform (GCP: Cloud Run), Terraform, Terragrunt, Docker, GitHub CI/CD, uv
Data & Backend: PostgreSQL, Supabase, Redis, Prisma, FastAPI, Next.js, n8n, SendGrid
Visualisation & Collaboration: Tableau, Jira, GitHub

Certification & Award

- PLUS Award | University of Bristol, United Kingdom | 17th March, 2026
- Certificate of Advanced Data Analytics Professional | Korea Data Agency, Korea | 17th Nov, 2023